Appendix to "Tools to Explain Neural Networks for Power System Dynamics"

A. Stiffness

Power-system dynamic models are difficult to simulate primarily because they are *stiff*, meaning that they contain modes evolving on widely separated time scales. To make this precise, consider the linearized differential dynamics

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x},$$

where $\mathbf{A} = \partial f / \partial \mathbf{x}$ is the Jacobian of the differential part of the model. Let λ_j denote the eigenvalues of $\mathbf{A}$.

For stable modes, the decay of each component is governed by $\Re(\lambda_j)$. We therefore define the modal decay time scale as

$$\tau_j = \frac{1}{|\Re(\lambda_j)|}.$$

If these decay time scales differ by several orders of magnitude, the system is stiff. A convenient measure is the stiffness ratio

$$r = \frac{\max_j |\Re(\lambda_j)|}{\min_j |\Re(\lambda_j)|},$$

where the minimum and maximum are taken over the modes with nonzero real part. Equivalently,

$$r = \frac{\tau_{\max}}{\tau_{\min}}.$$

In power-system component models, stiffness arises because electrical, control, and electromechanical states evolve on very different time scales. Fast current or voltage-control modes may decay in sub-milliseconds, while mechanical or outer-control modes may evolve over tens of milliseconds or even seconds. This separation is the main reason why accurate time-domain simulation becomes computationally demanding.

B. Numerical Stability

Numerical stability is conveniently understood by applying a time-stepping scheme to the scalar test equation

$$\dot{u} = \lambda u, \qquad \Re(\lambda) < 0.$$

After one time step of size h, a one-step integration method produces

$$u_{k+1} = R(h\lambda)\, u_k, \tag{1}$$

where $R(z)$ is the stability function of the method.

Stability requires

$$|R(h\lambda)| < 1$$

for the relevant eigenvalues of the system Jacobian. For explicit Euler,

$$R(z) = 1 + z,$$

so stability requires the time step to resolve the fastest decaying mode:

$$h \ll \frac{1}{\max_j |\Re(\lambda_j)|}.$$

Thus, one very fast mode can force a prohibitively small step size even if the quantities of interest evolve much more slowly.

Implicit methods have much larger stability regions and are therefore preferred for stiff systems. However, they require the solution of nonlinear algebraic systems at every time step. In large-scale power-system simulations, this creates a trade-off: explicit methods are cheap per step but constrained by stability, whereas implicit methods are stable for larger steps but expensive because each step requires iterative solves.

This same multiscale structure reappears in the learning problem considered in this paper. There, stiffness no longer appears as a restriction on the integration step size, but as an imbalance in the rates at which different components of the training error decay.

C. Synchronous Generator Models

The second-order model, referred to as the *2D scenario*, captures essential rotor dynamics governed by the power balance at the synchronous machine (SM) rotor. The states are the rotor angle δ and speed deviation ω, with dynamics:

$$\dot{\delta} = \omega, \tag{2a}$$

$$\dot{\omega} = \frac{\omega_b}{2H}\left(P_m - P_e - D\,\omega\right), \tag{2b}$$

where ω_b is the base electrical angular frequency, H is the inertia constant, P_m and P_e are the mechanical and electrical power, respectively, and D is the damping coefficient. Assuming an infinite bus, the electrical power is $P_e = V_1 V_2 \sin(\delta)$, where V_1 is the machine terminal voltage and V_2 the infinite-bus voltage magnitude.

a) 4D Scenario.: The fourth-order model extends the 2D model by including transient internal voltage dynamics (e'_d, e'_q):

$$\dot{e'_d} = \frac{1}{T'_{q0}}\left(-e'_d + i_q(x_q - x'_q)\right), \tag{3a}$$

$$\dot{e'_q} = \frac{1}{T'_{d0}}\left(-e'_q + i_d(x_d - x'_d) + e_{fd}\right), \tag{3b}$$

$$P_e = e'_d i_d + e'_q i_q + i_d i_q (x'_q - x'_d). \tag{3c}$$

Here, T'_{d0} and T'_{q0} are transient open-circuit time constants, x_d, x_q and x'_d, x'_q are synchronous and transient reactances, i_d, i_q are stator currents, and e_{fd} is the field voltage.

TABLE I: Parameters used for synchronous generator modeling.

Parameter	Value
Synchronous Machine Parameters	
Damping coefficient, D	2
Inertia constant, H [s]	5.06
Base angular frequency, ω_b [rad/s]	314.159
Transient time constants, T'_{d0}, T'_{q0} [s]	4.75, 1.6
Subtransient time constants, T''_{d0}, T''_{q0} [s]	0.03, 0.05
Direct-axis reactances, x_d, x'_d, x''_d [p.u.]	1.25, 0.232, 0.20
Quadrature-axis reactances, x_q, x'_q, x''_q [p.u.]	1.22, 0.715, 0.20
Field voltage, e_{fd} [p.u.]	1
Infinite bus voltage, V_2 [p.u.]	1

TABLE II: Input-domain variables for synchronous generator simulations.

Variable	Domain
Rotor States	
Rotor angle, δ [rad]	$[-0.05,\ 0.05]$
Rotor speed deviation, ω [rad/s]	$[-0.05,\ 0.05]$
Transient and Subtransient Voltages	
Transient direct-axis voltage, e'_d [p.u.]	0 (fixed)
Transient quadrature-axis voltage, e'_q [p.u.]	$[0.99,\ 1.02]$
Subtransient direct-axis voltage, e''_d [p.u.]	$[0.99,\ 1.02]$
Subtransient quadrature-axis voltage, e''_q [p.u.]	$[0.99,\ 1.02]$
Mechanical / Field Inputs	
Mechanical power, P_m [p.u.]	0.7048 (fixed)
Field voltage, e_{fd} [p.u.]	1.08 (fixed)

b) 6D Scenario.: The sixth-order model further includes subtransient internal voltage dynamics (e''_d, e''_q):

$$\dot{e''_d} = \frac{1}{T''_{q0}}\left(e'_d - e''_d - i_q(x'_q - x''_q)\right), \tag{4a}$$

$$\dot{e''_q} = \frac{1}{T''_{d0}}\left(e'_q - e''_q - i_d(x'_d - x''_d)\right). \tag{4b}$$

1) Input Domain: The input domain Ω_d defines the space from which initial conditions for the simulations are drawn. Since the present work considers only the synchronous machine dynamics described in the 2D, 4D, and 6D scenarios, the input domain must include initial values for the states involved in these models, namely: rotor angle δ, rotor speed deviation ω, transient voltages (e'_d, e'_q), and, when applicable, the subtransient voltages (e''_d, e''_q). The sampling ranges for these variables are summarized in Table II. A total of 400 initial conditions were generated via Latin Hypercube Sampling (LHS) over this domain. The same domain and sampling strategy were used to initialize collocation points.

The rotor angle δ and speed deviation ω define the electromechanical initial operating point of the synchronous generator. The transient and subtransient voltages $(e'_d, e'_q, e''_d, e''_q)$ represent the internal electromagnetic states of the machine and determine short-term voltage dynamics under the 4D and 6D models. Fixed values for the field voltage e_{fd} and mechanical input P_m correspond to steady operating conditions of the generator in the absence of explicit excitation-system or governor dynamics. The selected intervals capture a realistic but sufficiently wide range of perturbations around nominal operating points, enabling robust evaluation of the learned dynamics across multiple machine orders.

D. Second-Order Electromechanical Oscillator with Fast Voltage Dynamics

To study the effect of one dominant fast electrical mode, we consider the modified SMIB model

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & H & 0 \\ 0 & 0 & \tau \end{bmatrix} \frac{d}{dt}\begin{bmatrix} \delta \\ \omega \\ V \end{bmatrix} = \begin{bmatrix} \omega \\ P_m - EV\sin(\delta) + P_L(t) - D\omega \\ V_{\text{ref}} - V + k\sin(\delta) \end{bmatrix}. \tag{5}$$

Equivalently, in explicit form,

$$\dot{\delta} = \omega, \tag{6}$$

$$\dot{\omega} = \frac{1}{H}\Big(P_m - EV\sin(\delta) + P_L(t) - D\omega\Big), \tag{7}$$

$$\dot{V} = \frac{1}{\tau}\Big(V_{\text{ref}} - V + k\sin(\delta)\Big). \tag{8}$$

The parameters used in this study are

$$H = 10\ \text{s}, \qquad D = 1\ \text{p.u.}/(\text{rad/s}), \qquad P_m = 1\ \text{p.u.},$$
$$E = 1\ \text{p.u.}, \qquad \tau = 0.01\ \text{s},$$

with disturbance

$$P_L(t) = 0.5\sin(10t),$$

and initial conditions

$$\delta(0) = 0, \qquad \omega(0) = 0, \qquad V(0) = 1.$$

a) Local time-scale estimate.: Because the forcing term $P_L(t)$ is time-varying, the system is not analyzed through a single global equilibrium. Instead, the stiffness estimate used in the paper is obtained from the Jacobian of the *frozen* dynamics around a nominal operating point $(\delta^\star, \omega^\star, V^\star)$. For the explicit form above, the Jacobian is

$$\mathbf{A}(\delta^\star, \omega^\star, V^\star) = \begin{bmatrix} 0 & 1 & 0 \\ -\dfrac{EV^\star\cos(\delta^\star)}{H} & -\dfrac{D}{H} & -\dfrac{E\sin(\delta^\star)}{H} \\ \dfrac{k\cos(\delta^\star)}{\tau} & 0 & -\dfrac{1}{\tau} \end{bmatrix}.$$

For a nominal operating point with small angle deviation and $V^\star \approx 1$, this becomes approximately

$$\mathbf{A} \approx \begin{bmatrix} 0 & 1 & 0 \\ -\dfrac{E}{H} & -\dfrac{D}{H} & 0 \\ \dfrac{k}{\tau} & 0 & -\dfrac{1}{\tau} \end{bmatrix}.$$

This form shows the key structure directly:

- the voltage state contributes a fast electrical mode with decay rate of order $1/\tau$;
- the (δ, ω) subsystem contributes the slower electromechanical dynamics, whose decay rate is of order $D/(2H)$ in the lightly damped regime.

Accordingly, the dominant decay rates are approximately

$$|\Re(\lambda_{\text{fast}})| \approx \frac{1}{\tau} = 100, \qquad |\Re(\lambda_{\text{slow}})| \approx \frac{D}{2H} = 0.05.$$

The corresponding stiffness ratio is therefore

$$r_{\text{SMIB}} \approx \frac{|\Re(\lambda_{\text{fast}})|}{|\Re(\lambda_{\text{slow}})|} = \frac{1/\tau}{D/(2H)} = \frac{2H}{D\tau} \approx 2 \times 10^3.$$

This estimate shows that the modified SMIB benchmark is stiff, but in a comparatively simple way: the stiffness is dominated by essentially one fast electrical direction, while the remaining dynamics evolve on a much slower electromechanical time scale.

E. Grid-Following Inverter Model Description

The grid-following inverter considered in this work combines a phase-locked loop (PLL), an outer active/reactive power controller, an inner current controller, an RL electrical interface, and first-order measurement filters. All quantities are expressed in the synchronous dq frame aligned with the PLL angle.

a) State, input, and parameter vectors.: The eleven-dimensional state vector is

$$\mathbf{x} = \left[\xi_{\text{pll}},\, \theta_{\text{pll}},\, i_d,\, i_q,\, \xi_{i_d},\, \xi_{i_q},\, \xi_P,\, \xi_Q,\, P_{\text{filt}},\, Q_{\text{filt}},\, V_{\text{filt}}\right]^{\mathsf{T}}.$$

The input vector is

$$\mathbf{u} = \left[P_{\text{ref}},\, Q_{\text{ref}},\, V_{\text{ref}},\, \omega_{\text{ref}},\, v_{ga},\, v_{gb},\, v_{dc}\right]^{\mathsf{T}},$$

and the parameter vector collects the control gains and electrical parameters,

$$\mathbf{p} = \left[k_{p_{dp}},\, k_{i_{dp}},\, k_{p_{dq}},\, k_{i_{dq}},\, k_{p_{id}},\, k_{i_{id}},\, k_{p_{iq}},\, k_{i_{iq}},\, k_{p_{\text{pll}}},\, k_{i_{\text{pll}}},\, R,\, L,\, \omega_{\text{filt}}\right]^{\mathsf{T}}.$$

b) Synchronous-frame voltage transformation.: The PLL angle defines the synchronous frame:

$$v_{gd} = v_{ga}\cos\theta_{\text{pll}} + v_{gb}\sin\theta_{\text{pll}}, \tag{9}$$

$$v_{gq} = -\,v_{ga}\sin\theta_{\text{pll}} + v_{gb}\cos\theta_{\text{pll}}. \tag{10}$$

c) Phase-locked loop.: The PLL regulates the quadrature voltage component v_{gq} to zero through

$$\omega_{\text{pll}} = k_{p_{\text{pll}}}\, v_{gq} + k_{i_{\text{pll}}}\, \xi_{\text{pll}}, \qquad \omega = \omega_{\text{ref}} + \omega_{\text{pll}},$$

with state equations

$$\dot{\xi}_{\text{pll}} = v_{gq}, \qquad \dot{\theta}_{\text{pll}} = \omega.$$

d) Outer active and reactive power control.: The power-tracking errors are

$$dP = P_{\text{ref}} - P_{\text{filt}}, \qquad dQ = -Q_{\text{ref}} + Q_{\text{filt}}.$$

These generate current references through PI controllers:

$$i_{d,\text{ref}} = k_{p_{dp}}\, dP + k_{i_{dp}}\, \xi_P, \tag{11}$$

$$i_{q,\text{ref}} = k_{p_{dq}}\, dQ + k_{i_{dq}}\, \xi_Q, \tag{12}$$

with

$$\dot{\xi}_P = dP, \qquad \dot{\xi}_Q = dQ.$$

e) Inner current control and electrical plant.: The current controllers produce modulation voltages

$$v_{md} = k_{p_{id}}(i_{d,\text{ref}} - i_d) + k_{i_{id}}\, \xi_{i_d} - \frac{\omega}{\omega_{\text{ref}}} L\, i_q, \tag{13}$$

$$v_{mq} = k_{p_{iq}}(i_{q,\text{ref}} - i_q) + k_{i_{iq}}\, \xi_{i_q} + \frac{\omega}{\omega_{\text{ref}}} L\, i_d, \tag{14}$$

with integrator dynamics

$$\dot{\xi}_{i_d} = i_{d,\text{ref}} - i_d, \qquad \dot{\xi}_{i_q} = i_{q,\text{ref}} - i_q.$$

The converter-side RL network evolves according to

$$\dot{i}_d = -\frac{R}{L} i_d + \omega i_q + \frac{1}{L}(v_{md} - v_{gd}), \tag{15}$$

$$\dot{i}_q = -\frac{R}{L} i_q - \omega i_d + \frac{1}{L}(v_{mq} - v_{gq}). \tag{16}$$

f) Power and measurement filters.: The instantaneous powers are

$$P = v_{md} i_d + v_{mq} i_q, \qquad Q = v_{mq} i_d - v_{md} i_q,$$

and the converter-voltage magnitude is

$$v_m = \sqrt{v_{md}^2 + v_{mq}^2}.$$

These are filtered through first-order dynamics:

$$\dot{P}_{\text{filt}} = \omega_{\text{filt}}(P - P_{\text{filt}}), \tag{17}$$

$$\dot{Q}_{\text{filt}} = \omega_{\text{filt}}(Q - Q_{\text{filt}}), \tag{18}$$

$$\dot{V}_{\text{filt}} = \omega_{\text{filt}}(v_m - V_{\text{filt}}). \tag{19}$$

Collecting all terms gives the nonlinear state-space model

$$\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}, \mathbf{p}).$$

g) Nominal operating point.: Solving $f(\mathbf{x}^\star, \mathbf{u}^\star, \mathbf{p}^\star) = 0$ numerically yields the nominal equilibrium

$$\mathbf{x}^\star \approx \begin{bmatrix} -0.3491 \\ 0 \\ 0.9902 \\ 0 \\ 0.00202 \\ 0 \\ 0.01980 \\ 0 \\ 1.0000 \\ 0 \\ 1.0099 \end{bmatrix},$$

corresponding to

$$(\xi_{\text{pll}}, \theta_{\text{pll}}, i_d, i_q, \xi_{i_d}, \xi_{i_q}, \xi_P, \xi_Q, P_{\text{filt}}, Q_{\text{filt}}, V_{\text{filt}}).$$

h) Eigenvalue groups and time scales.: Linearizing the model around $(\mathbf{x}^\star, \mathbf{u}^\star, \mathbf{p}^\star)$ gives a Jacobian with 11 eigenvalues in total. Since complex eigenvalues occur in conjugate pairs, it is more convenient to report them as *eigenvalue groups*. Grouping conjugate pairs together, the spectrum is

$$\{-2552.29,\, -2500.28,\, -622.95,\, -619.97,\, -15.00 \pm 25.98j,\, -25.00,\, -2$$

Thus, the table below reports eight eigenvalue groups, which together account for all 11 eigenvalues of the Jacobian.

TABLE III: Dominant eigenvalue groups and associated decay time scales for the linearized inverter model. Complex-conjugate pairs are reported in one row.

Mode Group	Eigenvalue(s)	Decay time scale [s]	Dominant Variables
1	-2552.29	3.92×10^{-4}	$i_d,\ i_q,\ \xi_{i_d},\ \xi_{i_q}$
2	-2500.28	4.00×10^{-4}	$i_d,\ i_q,\ \xi_{i_d},\ \xi_{i_q}$
3	-622.95	1.61×10^{-3}	$i_d,\ i_q,\ \xi_{i_d},\ \xi_{i_q}$
4	-619.97	1.61×10^{-3}	$i_d,\ i_q,\ \xi_{i_d},\ \xi_{i_q}$
5	$-15.00 \pm 25.98j$	6.67×10^{-2}	$\xi_{\mathrm{pll}},\ \theta_{\mathrm{pll}},\ \xi_P,\ \xi_Q$
6	-25.00	4.00×10^{-2}	$P_{\mathrm{filt}},\ Q_{\mathrm{filt}},\ V_{\mathrm{filt}}$
7	$-25.01 \pm 25.20j$	4.00×10^{-2}	$P_{\mathrm{filt}},\ Q_{\mathrm{filt}},\ V_{\mathrm{filt}}$
8	$-25.26 \pm 25.05j$	3.96×10^{-2}	$P_{\mathrm{filt}},\ Q_{\mathrm{filt}},\ V_{\mathrm{filt}}$

i) Interpretation.: The inverter exhibits three clearly separated dynamical layers:

- **Fast layer:** the inner current-control and electrical RL dynamics, with decay time scales between approximately 4×10^{-4} s and 1.6×10^{-3} s;
- **Intermediate layer:** the measurement filters, with time scales near 4×10^{-2} s;
- **Slow layer:** the PLL and outer power-control loop, with a decay time scale near 6.7×10^{-2} s.

Using the real parts of the fastest and slowest eigenvalues, the corresponding stiffness ratio is approximately

$$r_{\mathrm{inv}} = \frac{2552.29}{15.00} \approx 1.7 \times 10^2.$$

Compared with the modified SMIB example, the inverter has a smaller overall stiffness ratio, but its stiffness is distributed across several interacting fast directions rather than being dominated by a single fast mode. This multi-directional stiffness is what makes the inverter substantially harder to learn in the results section.

REFERENCES